

Aakash Kumar

Software Engineer — Backend Systems & Data Infrastructure

+91 7320048899 — aakashkumar94303@gmail.com — aakash-kr-7 — aakash-kumar-978671274

Backend engineer focused on data-intensive systems, including vector retrieval, async concurrency, relational schema design, and scalable application backends. Built and independently deployed a live multi-user platform from scratch with custom retrieval pipelines, normalized persistence layers, concurrent session orchestration, and push notification infrastructure at \$0 operational cost.

Education

B.Tech in Computer Science & Engineering — Bennett University 2023 – Present
CGPA: 9.19 SAT: 1530/1600

Projects

SOL — Distributed Messaging Platform with Persistent Memory Architecture aakash-kr-7/sol
FastAPI · Flutter · ChromaDB · SQLite · Firebase · FCM · Railway

- Vector Retrieval:** Built pair-scoped **ChromaDB** namespaces giving each user-persona relationship its own isolated retrieval space, supporting **millions of memory interactions** with zero cross-namespace leakage. Engineered a deterministic **5-factor ranking engine** (similarity, salience, strength, recency decay, unresolved-context boost) to reduce long-horizon conversational drift.
- Persistence Layer:** Designed a normalized schema across **19 relational tables** and **28 indexed query paths**, separating high-frequency interaction telemetry from long-term retrieval and behavioral analytics workloads.
- Concurrency & Orchestration:** Engineered async lifecycle infrastructure using semaphore-controlled worker pools, pair-level locks, timeout-aware scheduling, and executor-offloaded blocking I/O, supporting **thousands of concurrent sessions** without race conditions or event-loop starvation.
- Behavioral Parameterization:** Modeled **12 distinct personas** through **132 runtime coefficients** and tracked relationship evolution across **6 behavioral metrics** and **5 adaptive stages**, driving dynamic conversational behavior from structured state rather than free-form prompting.
- Deployment:** Independently operated the full backend stack including FastAPI, ChromaDB, async workers, SQLite, and FCM on free-tier infrastructure at **\$0 operational cost** with a live, functional multi-user system.

AURORA — Speech Emotion Recognition System aakash-kr-7/aurora
Python · MFCC Feature Engineering · CNN / BiLSTM / SVM

- Built a unified training pipeline across **10K+ audio samples** from RAVDESS, CREMA-D, and TESS, handling feature extraction, normalization, augmentation, and dataset harmonization end-to-end.
- Benchmarked SVM (73%), BiLSTM (82%), and CNN (95%) architectures. Selected CNN as the optimal model based on its ability to extract time-frequency patterns from 2D MFCC spectrograms.

AETHER — Full-Stack Wellness Application aakash-kr-7/Aether
Flutter · Firebase · Firestore

- Designed a Firestore backend with **8+ user-scoped collections** around a single ownership model, enforced per-user data isolation through custom security rules, and modeled time-series journal schemas built for future analytics pipelines.
- Managed **20+ asynchronous UI states** across 10+ network-bound screens with explicit loading, failure, and partial-render handling throughout, without relying on optimistic fallbacks.

Research

Paraphrase-Robust Fake News Detection Using Deep Learning and Hybrid Models aakash-kr-7/fake-news-detection-ML-DL
NLP · Sentence Embeddings · Ensemble Methods

- Built a paraphrase-aware misinformation detection pipeline over **33,294 paraphrase sets** from **190K+ claims**, specifically designed to expose model failures under linguistically varied inputs.
- Best configuration: weighted lexical-semantic ensemble achieving **0.7956 F1**, **0.9328 prediction consistency**, and **0.9375 original-paraphrase agreement**, outperforming standalone TF-IDF, MiniLM, MPNet, and Logistic Regression baselines.

Technical Skills

Languages	Python, SQL, Dart, Java, C++
Backend & Databases	FastAPI, Firebase, Firestore, SQLite, ChromaDB, REST APIs, Relational Database Design
System Design	Async Programming, Concurrency, Data Modeling, Vector Retrieval, Clean Architecture
ML & Data	Sentence Embeddings, NLP, TF-IDF, Deep Learning, Feature Engineering, Model Benchmarking
App Development	Flutter, Firebase Authentication, Firebase Cloud Messaging
Tools	Git, GitHub, Linux, Android Studio, VS Code

Achievements & Certifications

- CodeChef:** Peak rating **1730** (Div. 2, 3★) — Global Rank **55** in best contest; top 250 globally in **2 contests**, top 1000 in **7 contests**.
- Top 100 of 2000+ University Projects** — Bennett University (Aether, 2025)
- Certified Ethical Hacker (CEH)** — EC-Council Feb 2026 – Mar 2027